

Handheld Shear Force Feedback Controller with Double Roller Modules: A Demonstration

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Abstract

A handheld shear force controller equipped with two roller modules was developed. The roller modules can rotate without angular limitations, making continuous directional force feedback to the palm possible through symmetric and asymmetric rotation, as well as ultralow-frequency vibration. By utilizing these novel features of the controller, its potential to create immersive user experiences was demonstrated through five applications.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

Haptics, Controller

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1 Introduction

The growing field of haptic technology is gradually being integrated into various industries, making the daily lives of people more interactive and immersive. Particularly in entertainment experiences, vibrotactile feedback generated by voice coil vibration actuators has become widely recognized as an essential feature of recent

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Figure 1: The prototype of the handheld shear force feedback controller.

commercial controllers. However, these voice coil actuators typically possess the ability to actuate vibrations at more than 50 Hz, which stimulates mechanoreceptors but may not always replicate the nuances of natural haptic sensations. While voice coil actuators are particularly effective in conveying brief expressions, such as clicks, impacts, and fine textures, they have difficulty representing continuous forces, directional forces, and ultralow-frequency vibrations over several seconds. There remains significant potential for enhancing the immersive quality of entertainment interactions by exploring novel haptic feedback technologies.

This research aims to address the challenge of creating continuous directional force feedback, which is a limitation of vibration actuators. Although the asymmetric actuation of vibration actuators can generate an illusory directional sensation [7], the forces that can be produced are weak, and the sensation of vibration remains, resulting in limited applicability. By contrast, various studies have been conducted on motor-driven haptic devices to pursue realism, employing rotating motors attached to handheld controllers to apply physical force feedback directly to the hand of a user [1] [9]. In addition, several investigations explored weight-shifting mechanisms to provide continuous feedback on the weights of handheld objects [11] [4]. These motor-driven force feedback systems primarily focus on kinematic sensations by directly stimulating the muscles and tendons of the user.

Conversely, shear force feedback induced by skin deformation from horizontally moving tactors has been researched as a different haptic feedback method that does not directly stimulate muscles and tendons. Several studies have demonstrated that shear forces resulting from skin deformation can provide a variety of force feedback modalities, including directional force, weight, and physical rendering. This phenomenon has been validated through numerous investigations on the fingers [5] [10] [8], wrists [6], and forearms [3] of users. In addition, a controller featuring multiple sliders arranged at the contact point on the palm has been proposed, making the representation of the rotational inertia and torque of virtual objects possible [2]. Most of these studies involve tactors with movable ends, which can pose challenges when expressing continuous forces.

In this research, a handheld controller equipped with two roller modules was developed that can rotate without angular limitations (figure 1). This controller makes independent control of positive and negative rotational directions possible. Each roller module is designed to present shear forces to the palm of the user, making stable and continuous force feedback possible, as well as ultralow-frequency vibration presentation. The capability of expressing diverse user experiences (UXs) across various applications by controlling the two roller modules symmetrically and asymmetrically. The contributions of this demonstration paper include the following.

- Continuous force feedback in the same direction is made possible by the unlimited rotation mechanism.
- Various directional forces are expressed through the synchronized operation of the two roller modules
- Ultralow-frequency vibration representation is achieved through fine reciprocal control of shear forces
- A shear force controller is proposed that realizes the above features.

2 Prototype

2.1 Controller Specifications

The prototype is a single handheld controller designed to generate shear forces on the palm of the user. It is equipped with two roller modules (figure 2). It measures 160 mm in length and weighs 286 g without a tracking device. The controller features buttons positioned for easy access by the index finger and thumb. It also includes an adjustable strap that enables users to experience force feedback while keeping their palms open, thereby preventing the

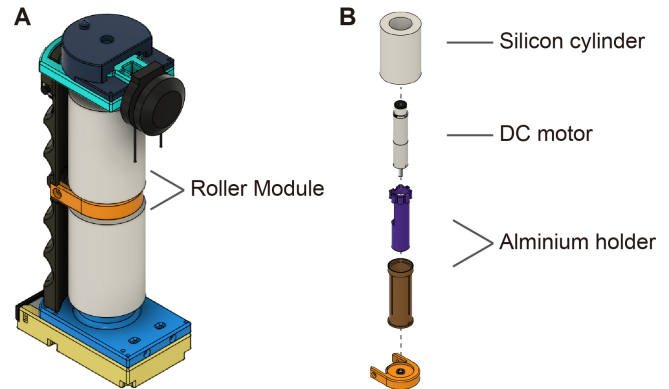


Figure 2: A: The prototype equipped with two roller modules. B: Components of the roller module.

controller from being dropped during the application of strong force feedback. A tracking device can be mounted on the top of the controller for positional tracking.

2.2 Roller Modules

The roller module primarily consists of a cylinder and a motor. The cylinder has a diameter of 34 mm and a length of 50 mm, providing a sufficient contact area with the palm when gripped by the user. The outer layer of the cylinder is made of 7.5-mm-thick silicone with a Shore A hardness of 50, ensuring a secure fit against the palm and efficient transmission of shear force feedback. The cylinder is connected to a Maxon DCX12L 35:1 motor, which can be actuated through either position control or torque control. The maximum continuous rating on the surface of the cylinder is 6.5 N. The motor holder is constructed from aluminum, making for efficient heat dissipation possible during the application of continuous torque. This module possesses sufficient backdrivability to release torque quality from the palm and return to an unloaded state when the voltage to the motor stops.

2.3 Unlimited Rotation

Although many prior studies have employed shear tactors with movable ends, the proposed prototype features roller modules that have no restrictions on their rotation angles. In the case of tactors with movable ends, when the grip of the user weakened and the tactor begins to spin freely, the range of motion of the tactor changes, preventing the ideal presentation of shear forces. Specifically, if the tactor reaches its movable end, it can no longer provide force in that direction. In contrast, the unlimited rotation design of the prototype eliminates the movable end, making stable and continuous force feedback possible in one direction. This is possible as long as the user maintains a proper grip on the device, even after free spinning occurs during use.

2.4 Actuation

By synchronizing the continuous actuation of the two roller modules with an unlimited rotation mechanism, a diverse range of expressions becomes possible (figure 3 B). Moreover, the precise

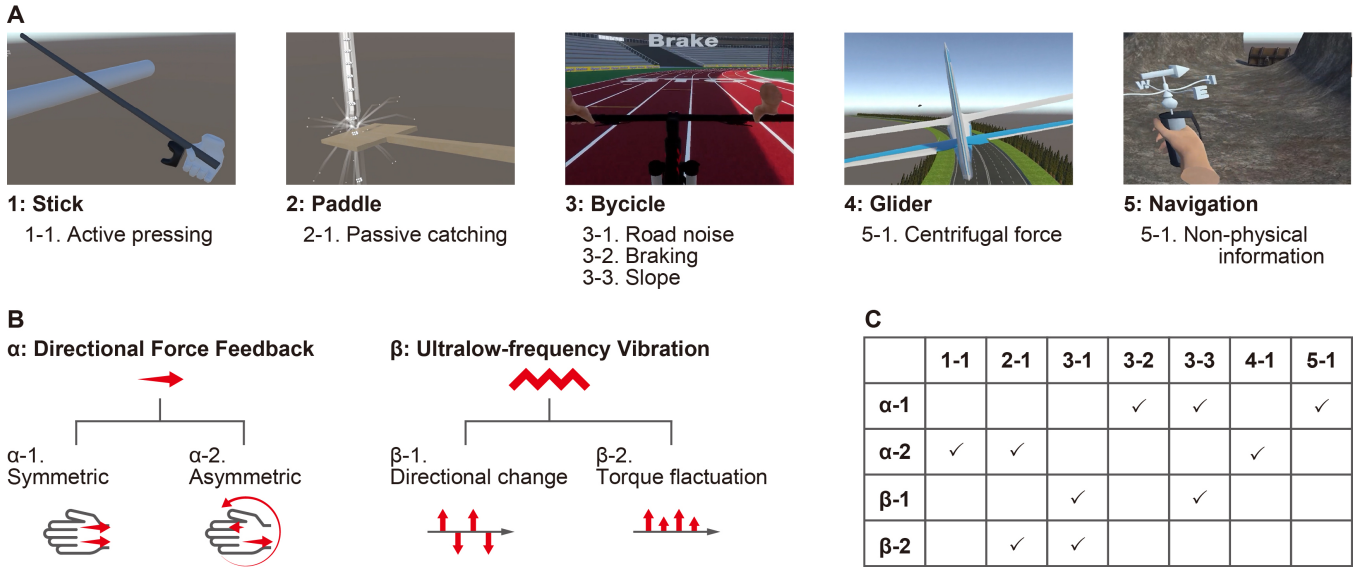


Figure 3: A: The list of the application and the UX. B: Actuation variations of the roller module. C: The matrix of the relation between the UX and the actuation.

control of the reciprocal rotation allows for the presentation of ultralow-frequency vibrations. The UX of this prototype, including its various applications, is explained in the following application section.

Synchronized Actuation. When the two roller modules are driven in synchronization, their actuation methods can be classified as symmetric or asymmetric. When both rollers rotate in the same direction, the user perceives a directional force. Conversely, when the two rollers rotate in opposite directions, the user experiences a rotational force.

Ultralow-Frequency Vibration. By rotating the rollers finely, it is possible to generate ultralow-frequency vibrations at approximately 10 Hz, a range that is typically unattainable using voice coil actuators. Vibrations can be produced by switching the positive and negative rotations of the rollers or by varying the torque in a single direction, making it possible to differentiate their use within the UX. This makes the representation of a solid vibration possible, rather than a collection of high-frequency vibrations that voice coil actuators typically achieve, through a single reciprocal motion.

3 Application

Five applications that utilize four distinct driving patterns are proposed (figure 3 A, C). These include continuous directional force feedback driven by the two rollers in both symmetric and asymmetric modes. In addition, ultralow-frequency vibration sensations are included through directional switching and variations in torque.

3.1 Stick

The user presses a stick held in the right hand against another object. The device provides continuous and directional force feedback when the stick is pressed against the object, simulating the force

applied by the stick on the palm of the user. During this interaction, the two roller modules of the device rotate in opposite directions, creating an illusory sensation of a rotational moment for the user. When the stick is pressed beyond a certain angle, it appears to pass through the object. Simultaneously, the force from the device is released, enabling the user to feel the sensation of the object being cut.

3.2 Paddle

The user catches a multitude of small objects that simulate falling water streams with a paddle-like object held in the right hand of the user. The device provides the force feedback that is expected to return to the palm of the user as the paddle is pushed downward. This feedback is achieved by the two roller modules of the device rotating in opposite directions. The control values of the device are based on the physical parameters calculated from the collision detection between the paddle and the small objects. By utilizing these control values in the torque output of the device, the ultralow-frequency vibrations resulting from the torque fluctuations represent the randomness of the water flow.

3.3 Bicycle

In this application, the controller is used to simulate the handlebars of a bicycle, expressing four elements: road noise, braking, slopes, and floating sensation.

Road Noise. The subtle vibrations experienced from the road surface while riding a bicycle are presented as ultralow-frequency vibrations at 10 Hz. The randomness of the road surface is represented by multiplying random values by the directional changes and torque fluctuations.

Braking. While riding, the brake is activated by pressing the button for the index finger on the controller. During braking, the sensation of being thrust forward because of inertia is simulated by rapidly rotating both roller modules in the opposite direction of the movement of the bicycle.

Slopes. Continuous directional forces are presented by both roller modules operating in the same direction to simulate the slope of the surface. When climbing uphill, the user experiences a sensation of being pushed back, which is represented by a rotation weighted in the opposite direction of travel. Conversely, when descending, the sensation of falling is conveyed through a rotation weighted in the direction of travel. Furthermore, the ultralow-frequency vibration of road noise is incorporated, with the minimized randomness in directional changes. This results in asymmetric vibrations in which a strong torque is presented on one side and weak torque on the other.

Grounded and Floating. While in motion, the device continuously provides haptic and auditory feedback from the resistance and impacts of the road surface. To simulate a state of floating in mid-air, the haptic and auditory feedback is momentarily interrupted, creating the illusion of being airborne.

3.4 Glider

The centrifugal force experienced while turning on a glider is simulated by varying the driving intensity and direction of the two roller modules. For example, while driving the two roller modules in the direction of glider movement, increasing the rotation of the right roller module simulates the centrifugal force directed to the right when the glider turns left. In addition, moving the two roller modules in opposite directions makes it possible to represent of lateral turns.

3.5 Navigation

The proposed controller can convey not only the physical forces experienced by the body, such as inertia and centrifugal force, but also non-physical information to the user. This application aims to assist navigation in environments where visual information is unavailable, such as dark caves. The directional force feedback derived from the rotation direction and strength of the roller modules can convey the correct direction to the user.

4 LIMITATION & FUTURE WORKS

The controller proposed is an initial prototype to validate the concept of unlimited rotation, and there remains significant room for improvement. Feedback from users of the demonstration indicated concerns about the mismatch between the sensation of rotation and the UX. This issue can be addressed by improving the shape and material of the cylinder through user testing. In addition, the size of the motors in the rotational mechanism presents a challenge for future integration into a compact controller. In this regard, it is necessary to investigate the quality of the UX that can be achieved while reducing the motor torque.

5 CONCLUSION

A handheld shear force controller aimed at achieving continuous force feedback on the hand of a user was proposed. This innovative controller is equipped with two roller modules that can rotate without any angular limitations, making a versatile range of interactions possible.

Five applications were presented that utilize the two roller modules in different ways, demonstrating the potential of this technology in various contexts. These applications illustrate how the controller can enhance the UX by providing realistic and immersive feedback.

Furthermore, these applications will be demonstrated at the conference, highlighting the practical implications and effectiveness of the proposed controller. This research contributes to the advancement of haptic technology and its applications.

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